PROJECT REPORT

IoT-Based Weather and Atmospheric Pressure Monitoring Station

A simulation-based Internet of Things project using ESP32, DHT22, BMP180, Wokwi and ThingSpeak

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DETAIL	INFORMATION
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PROJECT LINKS

Wokwi Simulation: Open interactive simulation

ThingSpeak Dashboard: View cloud channel

Abstract

This project presents a low-cost IoT-based weather and atmospheric pressure monitoring system implemented as an interactive simulation. An ESP32 microcontroller reads temperature and relative humidity from a DHT22 sensor and atmospheric pressure from a BMP180 sensor. The readings are transmitted over the Wokwi virtual Wi-Fi network to a ThingSpeak cloud channel, where they are stored and visualized as time-series graphs. The simulation allows users to change virtual sensor values and observe how the cloud dashboard responds. The work demonstrates sensor interfacing, embedded programming, wireless communication and cloud-based IoT data visualization. Since the system is simulated in Wokwi, its readings are test values rather than measurements of actual outdoor weather.

1. Introduction

The Internet of Things (IoT) connects physical or simulated devices to networks so that data can be collected, transmitted and analyzed. Environmental monitoring is a useful IoT application because parameters such as temperature, humidity and atmospheric pressure can be observed remotely and over time.

This project uses an ESP32 as the central controller. A DHT22 sensor supplies temperature and relative humidity data, while a BMP180 sensor supplies barometric pressure data. The ESP32 sends these readings to ThingSpeak, a cloud service that provides channel-based data storage and visualization. Wokwi is used to simulate the circuit and sensor behaviour, making the project accessible without physical components.

2. Objectives and Scope

- Interface a DHT22 temperature/humidity sensor and a BMP180 pressure sensor with an ESP32.
- Acquire and validate sensor readings using Arduino C++ firmware.
- Connect the ESP32 simulation to Wi-Fi.
- Upload temperature, humidity and pressure values to ThingSpeak.
- Visualize uploaded data through a public cloud dashboard.
- Demonstrate interactive testing by changing virtual sensor controls in Wokwi.

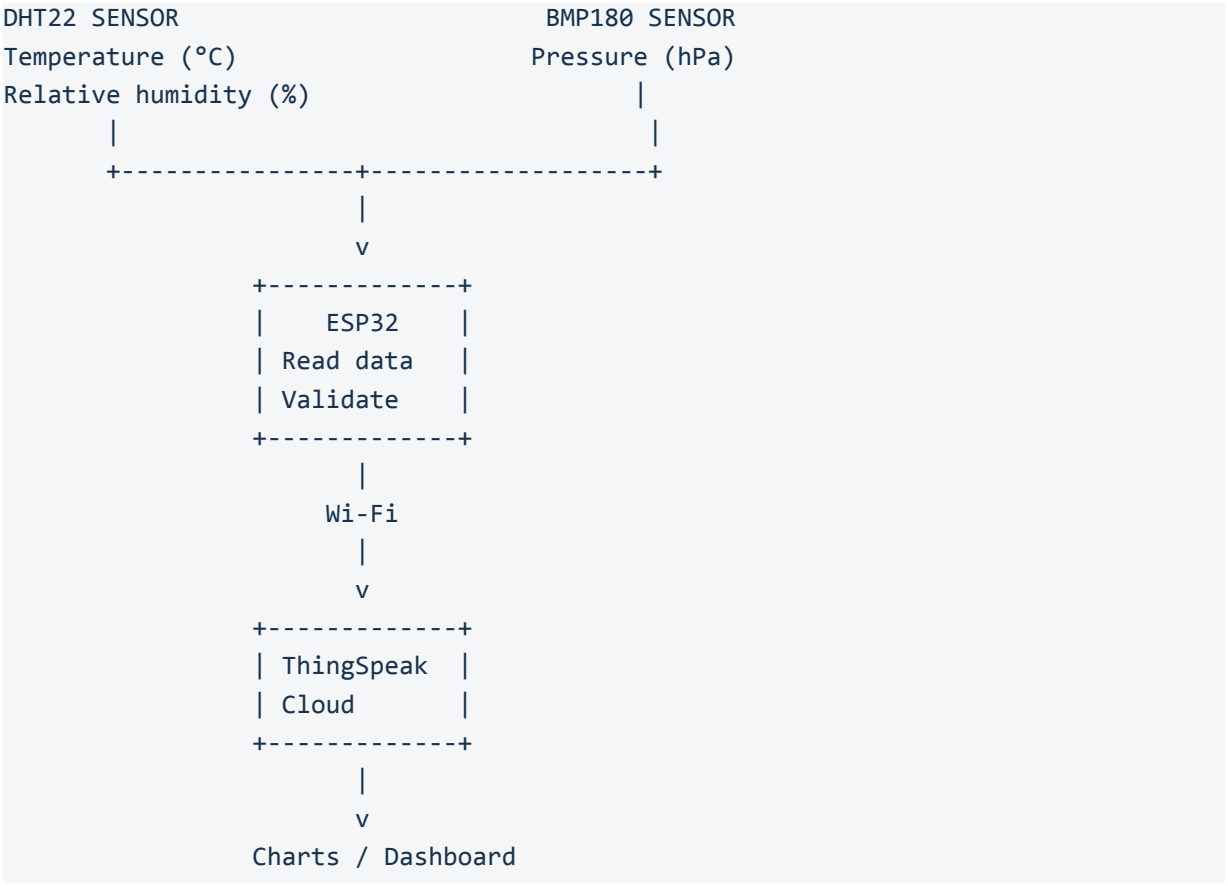
The scope is limited to a simulation-based prototype. It demonstrates the data flow and cloud integration, but does not provide verified outdoor weather measurements or a validated weather forecast.

3. System Requirements

CATEGORY	REQUIREMENT
Controller	ESP32 DevKit C (Wokwi simulation)
Sensors	DHT22 for temperature and humidity; BMP180 for atmospheric pressure
Development environment	Wokwi online simulator and Arduino C++
Libraries	Adafruit BMP085 Library; DHT sensor library for ESPx
Network	Wokwi virtual Wi-Fi network (Wokwi-GUEST)
Cloud platform	ThingSpeak channel configured with three data fields
Upload interval	Approximately 16 seconds between updates

4. System Architecture

The system follows a sensor-to-cloud architecture. Both sensors are connected to the ESP32. The ESP32 reads the sensor values, checks that the readings are valid, and sends them to ThingSpeak through a Wi-Fi connection.



4.1 Data Flow

1. Sensor acquisition: The ESP32 communicates with the DHT22 using a digital data connection and with the BMP180 over I²C.
2. Validation: The firmware checks sensor status and basic acceptable ranges before attempting an upload.
3. Transmission: The ESP32 submits the values to the ThingSpeak update endpoint over HTTPS.
4. Visualization: ThingSpeak stores accepted updates and plots each field on the channel dashboard.

The firmware uses an upload interval of approximately 16 seconds. This interval is intended to respect the typical minimum update interval of a free ThingSpeak channel.

5. Hardware Design and Connections

Although no physical circuit is assembled, the Wokwi diagram represents the following connections.

DEVICE	PIN / SIGNAL	ESP32 CONNECTION
DHT22	VCC	3V3
DHT22	SDA / DATA	GPIO 15
DHT22	GND	GND
BMP180	VCC	3V3
BMP180	GND	GND
BMP180	SDA	GPIO 21 (I²C SDA)
BMP180	SCL	GPIO 22 (I²C SCL)

The DHT22 data line is connected to GPIO 15 in the working simulation. The BMP180 uses the ESP32 I²C bus on GPIO 21 (SDA) and GPIO 22 (SCL). The DHT22 NC pin is left unconnected.

6. Software Design and Implementation

The firmware is written in Arduino C++ and runs on the simulated ESP32. Its main tasks are Wi-Fi connection management, sensor initialization, periodic sensor acquisition, input validation, pressure trend comparison and ThingSpeak upload.

SOFTWARE COMPONENT	RESPONSIBILITY
Wi-Fi setup	Connects to the Wokwi virtual access point using the network name Wokwi-GUEST.
I²C setup	Initializes the BMP180 bus using SDA GPIO 21 and SCL GPIO 22.
DHT22 setup	Initializes the DHT22 driver on GPIO 15.
Sensor reading	Reads temperature and humidity from DHT22 and pressure from BMP180.

Validation	Checks sensor status, finite numeric values and reasonable ranges.
Cloud upload	Sends the three values to ThingSpeak using an HTTPS POST request.

6.1 Sensor Data Mapping

VARIABLE	SOURCE	CHANNEL FIELD	UNIT
Temperature	DHT22	Field 1	°C
Humidity	DHT22	Field 2	%
Pressure	BMP180	Field 3	hPa

The BMP180 also provides a temperature measurement, but this project uses the DHT22 as the source of the reported temperature and humidity. The BMP180 is used for atmospheric pressure.

7. Cloud Integration

ThingSpeak is used as the cloud destination. The ESP32 sends the sensor values to the ThingSpeak update endpoint. Each accepted update is stored as a channel entry, and the channel's field charts display the values over time.

The channel is configured with Field 1 for Temperature (°C), Field 2 for Humidity (%) and Field 3 for Pressure (hPa). The channel is intended to be publicly viewable so visitors can inspect the demonstration data.

Wokwi simulation: https://wokwi.com/projects/479875277963187201

ThingSpeak channel: https://thingspeak.mathworks.com/channels/3567943

8. Testing and Observations

Testing is performed by running the Wokwi simulation, observing the ESP32 Serial Monitor and checking whether the ThingSpeak channel receives new entries. Virtual sensor controls allow the user to change the input values while the simulation is running.

TEST	PROCEDURE	EXPECTED RESULT
Wi-Fi connection	Start simulation and inspect Serial Monitor.	ESP32 connects to Wokwi-GUEST and reports status.
DHT22 reading	Read the virtual DHT22 sensor.	Temperature and humidity values are displayed.
BMP180 reading	Read the virtual BMP180 sensor.	Atmospheric pressure is displayed in hPa.
Cloud upload	Allow the upload interval to elapse.	A new ThingSpeak entry is created if accepted.

Input variation	Change virtual sensor values.	Subsequent readings and cloud charts reflect changed inputs.

8.1 Observed Simulation Behaviour

During sensor testing, the DHT22 returned a valid example reading of 24.00 °C and 40.00% relative humidity. The BMP180 produced a pressure reading of approximately 1013.27 hPa in an earlier test. These are example simulation observations, not fixed outputs or measurements of real weather.

The DHT22 initially produced timeout readings when connected to GPIO 4 in the test setup. Moving its data connection to GPIO 15 resolved the issue in this Wokwi project. The final documented wiring therefore uses GPIO 15 for DHT22 data.

Successful cloud transmission should be verified by checking that ThingSpeak displays a newly timestamped entry after the simulation sends an update. If the service rejects an update, the channel may not show a new entry.

9. Limitations

- The implementation is a Wokwi simulation and has not been validated with physical sensors.
- Sensor values are virtual test inputs and do not represent actual outdoor conditions.
- The public demo channel may receive readings from multiple visitors, so its data can be mixed with unrelated test activity.
- The pressure trend comparison is not a weather prediction model.
- A Write API Key embedded in public source code is visible to visitors and can be used by others to submit data to that channel. A disposable demo channel should be used; private account credentials and personal channel keys must not be published.

10. Future Enhancements

- Deploy the design on a physical ESP32 with real DHT22 and BMP180 sensors.
- Add sensors for rainfall, wind speed, light intensity or air quality.
- Store and analyze longer historical datasets.
- Develop a machine learning model for weather trend estimation using appropriate historical weather data.
- Add threshold-based alerts for unusual environmental conditions.
- Build a dedicated web or mobile dashboard for richer visualization.
- Use a secure backend service to protect write credentials when supporting public demonstrations.

11. Conclusion

The project demonstrates a complete simulation-based IoT data pipeline, from environmental sensor acquisition to cloud storage and visualization. The ESP32 reads temperature and humidity from the DHT22 and pressure from the BMP180, then transmits the values to ThingSpeak over Wi-Fi. Wokwi provides an interactive environment in which users can change sensor inputs and observe the resulting cloud data. The project provides practical experience in sensor interfacing, embedded programming, network communication and IoT cloud integration. A physical implementation and further validation would be required before using the system for real-world weather monitoring.

References

- Wokwi documentation: https://docs.wokwi.com/
- ThingSpeak platform: https://thingspeak.mathworks.com/
- Adafruit BMP085/BMP180 Library: https://github.com/adafruit/Adafruit-BMP085-Library
- DHT sensor library for ESPx, as listed in the Wokwi project libraries.